

Exchange rate and Production network

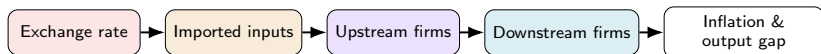
Bonn Macro Internal Seminar

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Motivation



Firms buy inputs mostly from *other domestic firms*, not just labor — a cost shock to one sector mechanically hits everyone downstream

Imported inputs are the norm (Chile: up to 19.5% in Manufacturing) — so an FX move is, for the firm, a domestic cost shock

Network models get this propagation right but ignore the exchange rate; open-economy models get the exchange rate right but ignore the network

Does accounting for production networks change the optimal exchange-rate regime?

Literature

Network side (closed economy): Rubbo (2024); La'O & Tahbaz-Salehi (2022); Pasten, Schoenle & Weber (2020); Ozdagli & Weber (2026); Ghassibe (2021)

Open-economy side (no network): Galí & Monacelli (2005); Fanelli & Straub (2021); Schmitt-Grohé & Uribe (2003); Gopinath & Itskhoki (2022); Amiti et al. (2019); Auer, Levchenko & Sauré (2019); Itskhoki & Mukhin (2023); Bianchi & Coulibaly (2025); Coulibaly (2023)

Open economy with network:

Gnocoato, Montes-Galdón & Stamato (2025, ECB); Kalemli-Özcan et al. (2025) — open economy *with* network (tariffs), fixed Taylor rule, no FX-regime choice¹

Qiu, Wang, Xu & Zanetti (2026, JME) — closest paper: derives the open-economy DC index directly extending Rubbo, WIOD 43-country calibration; but *static* (no NFA/persistence), no UIP, and no FX-regime choice

Gap: network + open economy + FX regime choice, together.

¹Details & what these papers cite for open-economy production networks: Appendix.

This Paper

Extend Rubbo (2024) → SOE: imported inputs, UIP and debt-elastic premium

Three literatures, none overlapping fully (network only; open economy only; network + open economy but no regime choice) — **first to put all three together**

Calibrate: Chile's real IO table, also Korea, Czechia

Ask: rank FX regimes? How does the network amplify foreign shocks? How do sector heterogeneity and network structure matter?

Result: Managed < Float $\ll$ Peg (welfare loss $10.96/25.15/90.45 \times 10^{-4}$)

Seven shocks, two kinds — keep this distinction throughout: **cost-push** (sector TFP A_i , import price PF) enters marginal cost directly and breaks divine coincidence; **demand** (foreign demand D^* , export price/ToT PX , risk-premium/UIP RP) moves $\tilde{y}_t$ without a first-order cost-push term of its own — but under a **fixed** exchange rate, the interest-rate response needed to defend $\hat{s}_t=0$ turns even a pure demand shock into a large output-gap cost (see Peg, throughout)

Households

Representative household chooses $\{C_t, L_t, B_t, B_t^*\}$ to maximize lifetime utility, subject to a budget constraint with a domestic bond B_t and a foreign bond B_t^* :

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \right]$$

subject to

$$P_t^C C_t + B_t + S_t B_t^* = W_t L_t + \Pi_t + (1+i_{t-1})B_{t-1} + (1+i^*)[1-\psi(\hat{b}_{t-1}^* - \bar{b}^*)]S_t B_{t-1}^*$$

Consumption is CES across home/foreign goods, Cobb-Douglas across domestic sectors:

$$C_t = \left[\omega^{1/\eta} (C_t^H)^{\frac{\eta-1}{\eta}} + (1-\omega)^{1/\eta} (C_t^F)^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}, \quad C_t^H = \prod_i (C_{it}^H)^{\beta_i^H}$$

Firms: Technology

Firms in sector i share Cobb-Douglas technology (labor, domestic inputs, imports) and price under monopolistic competition:

$$Y_{ift} = A_{it} L_{ift}^{\alpha_i} \prod_j X_{ijft}^{\Omega_{ij}^H} M_{ift}^{\Omega_i^F}$$
$$MC_{it} = \frac{W_t^{\alpha_i} \prod_j P_{jt}^{\Omega_{ij}^H} (S_t P_t^{F*})^{\Omega_i^F}}{A_{it}}$$

Two cost-push channels: MC_{it} moves with A_{it} (TFP) and S_t , the exchange rate (via the import cost $S_t P_t^{F*}$).

The theory holds for any number of sectors N ; the quantitative results use $N = 3$ (Resource, Manufacturing, Services), collapsed from Chile's 12-sector national IO table.

Firms: Price Setting

Calvo pricing through the network delivers the (vector) Phillips curve:

$$\boldsymbol{\pi}_t = \rho(I - \mathcal{V})\mathbb{E}_t\boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\boldsymbol{\chi}_t + \Gamma\Delta e_t$$

Same amplifier, two shocks: $\mathcal{B}, \Gamma$ are the *same* rigidity-adjusted operator $\tilde{A} \equiv \Delta(I - \Omega^H \Delta)^{-1}$, applied to labor share α vs. import share ω_F :

$$\mathcal{B} = \frac{\tilde{A}\alpha}{\kappa}, \quad \Gamma = \frac{\tilde{A}\omega_F}{\kappa}$$

where:

$\boldsymbol{\pi}_t, \tilde{y}_t$: infl., gap	$\boldsymbol{\chi}_t$: TFP cost-push	Δe_t : exch. rate chg.	κ : GE scalar
$\mathcal{B}, \Gamma$: PC slope, FX pass-thru	$\mathcal{V}$: TFP cost-push matrix	$\tilde{A}$: rigidity-adj. Leontief	

Firms: Production Network

The **Leontief inverse** sums direct and indirect linkages:

$$(I - \Omega^H)^{-1} = I + \Omega^H + (\Omega^H)^2 + \dots$$

applied to consumption shares β^H (Households) it gives:

$$\lambda_D^\top = (\beta^H)^\top (I - \Omega^H)^{-1}, \quad \mathcal{M} = (I - \Omega^H)^{-1} \omega_F, \quad \mathcal{M}^X = (I - \Omega^H)^{-1} \omega_X$$

where:

$\lambda_{D,i}$: **Domar weight** — sector i 's total weight in final demand, network-adjusted (can exceed 1 for downstream sectors)

$\mathcal{M}_i$: **Import centrality** — sector i 's total (direct + indirect) exposure to import costs

$\mathcal{M}_i^X$: **Export centrality** — sector i 's total (direct + indirect) exposure to export demand

Why left vs. right multiply: λ_D asks "how much of *final demand* ultimately flows through sector i ?" — a question about a row vector of weights pulled *backward* from consumption, hence $(\beta^H)^\top (\cdot)$ on the left. $\mathcal{M}, \mathcal{M}^X$ ask "how much of a *cost/demand shock at the border* reaches sector i ?" — a question about a column vector pushed *forward* through supply chains, hence $(\cdot) \omega_F$ on the right. Same Leontief inverse, opposite direction of the question

Foreign Sector

Law of one price and UIP with a debt-elastic NFA stabilizer $\psi(\cdot)$ and an exogenous risk-premium shock RP_t :

$$P_t^{FH} = S_t P_t^{F*}, \quad i_t = i^* + \mathbb{E}_t \Delta e_{t+1} + \psi(\hat{b}_t^*) + RP_t$$

NFA accumulates the trade balance:

$$\hat{b}_t^* = \frac{1 + i_{t-1}}{\Pi_t^C} \hat{b}_{t-1}^* + \frac{P_t^H \sum_i EX_{it} - P_t^{FH} IM_t}{P_t^C}, \quad EX_{it} = \kappa_{EX,i} D_t^* PX_t \left(\frac{P_{it}}{P_t^{FH}} \right)^{-\theta^*}$$

Two channels to the domestic economy:

$$\hat{s}_t + \hat{p}_t^{F*} \rightarrow MC_{it} \rightarrow \pi_{it} \quad (\text{cost-push}) \quad \hat{s}_t \rightarrow EX_{it} \rightarrow \tilde{y}_t \rightarrow \pi_{it} \quad (\text{demand})$$

where:

P_t^{F*} : foreign price of imports
 $\psi(\cdot)$: debt-elastic NFA stabilizer
 $\hat{s}_t \equiv \log(S_t/\bar{S})$: log exch. rate

RP_t : risk-premium shock
 $\hat{b}_t^*$: NFA (% GDP)
 $\Delta e_{t+1} \equiv \mathbb{E}_t \hat{s}_{t+1} - \hat{s}_t$: exp. depreciation

D_t^* : foreign demand shock
 $\kappa_{EX,i}$: export scale, sector i
 PX_t : export price/ToT shock
 θ^* : export demand elasticity
 (hats: log-deviation from steady state, throughout)

Monetary Policy Regimes

Free float

$$\hat{i}_t = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t$$

The exchange rate adjusts endogenously.

Hard peg

$$\hat{s}_t = 0$$

Adjustment occurs through domestic inflation and output.

Managed float

$$\hat{i}_t = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \hat{s}_t$$

The central bank partially stabilizes the exchange rate.

Equilibrium System

Phillips curve:

$$\pi_t = \rho(I - \nu)\mathbb{E}_t\pi_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \nu\chi_t + \Gamma\Delta e_t$$

IS curve:

$$\tilde{y}_t = \mathbb{E}_t\tilde{y}_{t+1} - \frac{1}{\gamma}(i_t - \mathbb{E}_t\pi_t^C - r_t^{\text{nat}}) - \frac{\eta(1 - \omega)}{\gamma}\mathbb{E}_t\Delta(\hat{s}_t + \hat{p}_t^{F*} - \hat{p}_t^H)$$

Foreign sector:

$$i_t = i^* + \mathbb{E}_t\Delta e_{t+1} + \psi(\hat{b}_t^*) + RP_t$$

Policy rule:

$$\hat{i}_t = \begin{cases} \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t & \text{Float} \\ \text{residual } (\hat{s}_t = 0) & \text{Peg} \\ \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \hat{s}_t & \text{Managed} \end{cases}$$

Not over-identified: law of one price ($P_t^{FH} = S_t P_t^{F*}$, Foreign Sector slide) is a *definition* substituted into χ_t /the IS curve's real-exchange-rate term, not an extra equilibrium condition — the system is N Phillips curves + IS + UIP + policy rule solving for N sectoral π_{it} , $\tilde{y}_t$, i_t , $\hat{s}_t$: equal counts, one equation per unknown.

Shock	A_1	A_2	A_3	PF	D^*	PX	RP
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s.d. (all)

1%

Calibration: Chile Data

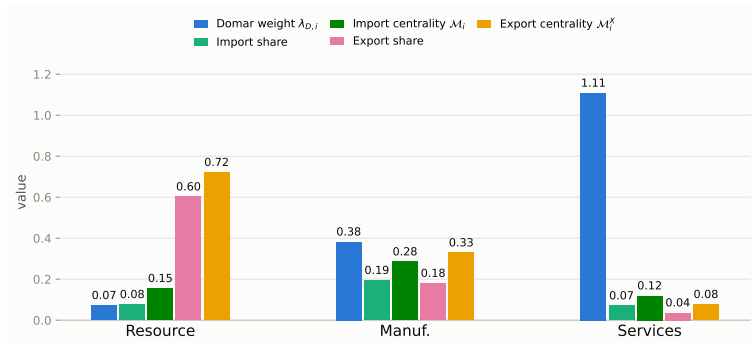
Param.	Description	Value	Source
$\alpha_1, \alpha_2, \alpha_3$	Labor share	0.503, 0.359, 0.604	Data: BCCh IO
$\Omega_1^F, \Omega_2^F, \Omega_3^F$	Import cost share	0.077, 0.195, 0.070	
$\beta_1^H, \beta_2^H, \beta_3^H$	HH consumption share	0.027, 0.229, 0.744	
Export share _{1,2,3}	Export share of output	0.602, 0.180, 0.036	
$\delta_1, \delta_2, \delta_3$	Calvo reset prob.	0.90, 0.31, 0.16	Lit.: Dhyne et al. (2005); N-S (2008)
ψ	Risk-premium elasticity	0.020	Key mechanism param. (see appendix)
ϕ_π	Taylor: inflation resp.	1.50	
ϕ_y	Taylor: output-gap resp.	0.50	
ϕ_s	Taylor: FX-smoothing resp.	0.30	

Ω^H (order: Res., Man., Serv.; source: BCCh IO table):

$$\Omega^H = \begin{pmatrix} 0.075 & 0.153 & 0.193 \\ 0.099 & 0.202 & 0.145 \\ 0.002 & 0.058 & 0.266 \end{pmatrix}$$

Standard literature values (not paper-specific; sources in Appendix): $\beta=0.99$, $\gamma=1.00$, $\varphi=2.00$, $\eta=1.50$, $\varepsilon=8.00$, $\theta^*=2.00$, $\beta^F, \text{tot}=0.10$.

Network Properties: Import & Export Exposure



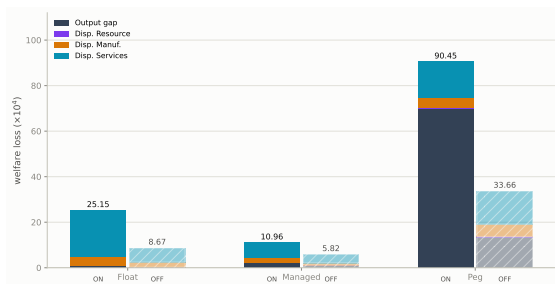
Reading the Robustness Slides: What Varies, What's Fixed

Every comparison slide that follows dials **one** object and holds the rest of the calibration fixed — but "fixed" doesn't always mean "untouched": some dials force a mechanical rebalancing elsewhere so cost shares still sum to one. This is the map.

Slide / dial	Varies	Held fixed / mechanically adjusted
Isolating the Network Channel; ϕ_s vs. network; Why Services?	cross-sector density ρ (off-diag. Ω^H)	own-input Ω_{ii}^H , imports Ω_i^F fixed; α_i (labor share) absorbs the change so shares sum to 1
Import / Export Openness	Ω_i^F (or export share) scale	everything else fixed at Chile baseline; α_i absorbs
Risk-Premium Volatility	$sd(\varepsilon^{RP})$ only	persistence $\rho_{RP}=0.80$, every other shock, all cost shares — nothing else moves
Korea / Czechia	entire calibration ($\Omega^H, \Omega^F, \beta^H$)	(all only functional-form parameters ($\gamma, \varphi, \varepsilon, \beta$) shared with Chile

Not yet swept: the debt-elastic risk-premium *elasticity* ψ itself (only its shock's *volatility* is, above) — flagged as the one remaining diagnostic on the "75% risk-premium share" result.

Welfare Ranking: Managed Float Dominates



Managed < Float ≪ Peg: $10.96/25.15/90.45 \ (\times 10^{-4})$

Float & Managed: almost entirely price dispersion, concentrated in Services

Peg: almost entirely output gap

Without the network: same ranking, but the network roughly **doubles-to-triples** every regime's loss

Units: $\mathcal{W} = \frac{1}{2}[(\gamma + \varphi)\text{Var}(\check{y}_t) + \sum_i \kappa_i \text{Var}(\pi_{it})]$ is a **per-period, consumption-equivalent fraction** (standard Rotemberg–Woodford quadratic-loss normalization) — Managed's 10.96×10^{-4} means an average per-period loss of $\approx 0.11\%$ of steady-state consumption, every period, forever, from stabilization policy alone (on top of any level effects, which this second-order

Counting the Cross-Sector Term (Rubbo Prop. 3)

Christian: *"I really need to count for that."* Rubbo's welfare formula has a **between-sector** piece, $\Phi_C + \sum_s \lambda_s \Phi_s$ (relative-price misallocation across sectors), that every number so far omits. Now built and computed, not just flagged.

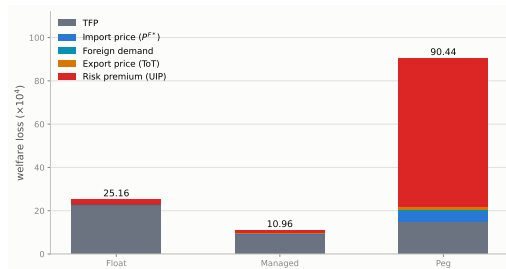
Closed form: both cross-sector aggregators here are **Cobb-Douglas** (household consumption across domestic sectors β^H ; firm input bundles Ω^H, Ω^F) — Cobb-Douglas has Allen-Uzawa elasticity of substitution *exactly* 1 by construction, so Φ_C, Φ_s collapse to a **weighted variance of the markup gap** $\mu_{it} = \log(P_{it}/MC_{it}) - \log \frac{\varepsilon}{\varepsilon-1}$ across sectors (only the outer C^H/C^F nest is genuine CES, $\eta=1.5$, used for the one domestic-vs-import pair). μ_{it} added as a pure reporting definition to the .mod files — changes nothing else.

	Float	Managed	Peg
Within-sector + output gap (headline)	25.47	10.17	102.05
Cross-sector $\Phi_C + \sum_s \lambda_s \Phi_s$ (new)	0.83	0.86	4.57
New total	26.31	11.03	106.62
% added	3.3%	8.4%	4.5%

Ranking unchanged: Managed < Float << Peg, same as before — the cross-sector term is real (not zero) but **second-order** relative to within-sector dispersion, because it's mechanically bounded by the *spread* across only 3 sectors' price gaps, not any single sector's own variance. **Caveat:**

domestic vs import cross elasticity uses η as a practical stand in for the true nested CES

What Drives Each Regime's Loss? Shock Decomposition



Peg: risk-premium/UIP $\rightarrow$ 76% of loss, mostly output gap

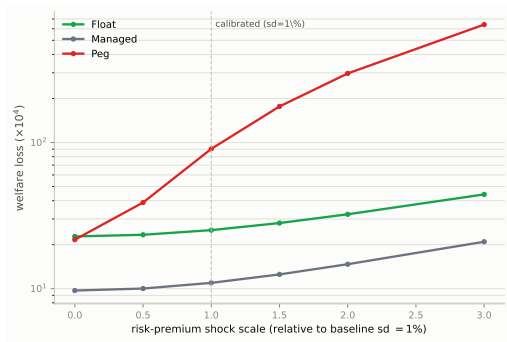
Float & Managed: TFP shocks dominate instead — risk-premium barely registers since the exchange rate (Float) or partial smoothing (Managed) absorbs it before it reaches output

Import price, foreign demand, and export price are second-order for every regime

How to read this: allocates *total welfare loss* (previous slide) across shocks using Dynare's own variance decomposition, exactly — %-of-variance-by-shock $\times$ that variable's total variance, summed with the same $\kappa_i/(\gamma+\varphi)$ weights as the welfare formula, so shock contributions add up

How Much Does Peg's Loss Depend on the Risk-Premium Shock?

Adam: "effects of the volatility of the exchange-rate processes." Risk-premium/UIP drives 76% of Peg's loss (previous slide) — calibrated at the **same 1% s.d.** as every other shock, no independent target. **Important, uncomfortable finding, stated plainly:**

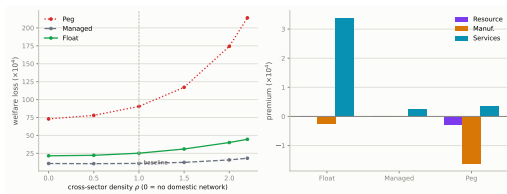


With the risk-premium shock switched off entirely, Peg is *not* worst — Peg $0.95 \times$ Float (essentially tied, Peg fractionally *better*). The entire "Peg is dominated" headline rests on this one shock existing, not on terms-of-trade, import pass-through, or any network channel

Turn it on at **half** the calibrated size and Peg is already $1.7 \times$ Float; at the **calibrated** size,

Isolating the Network Channel

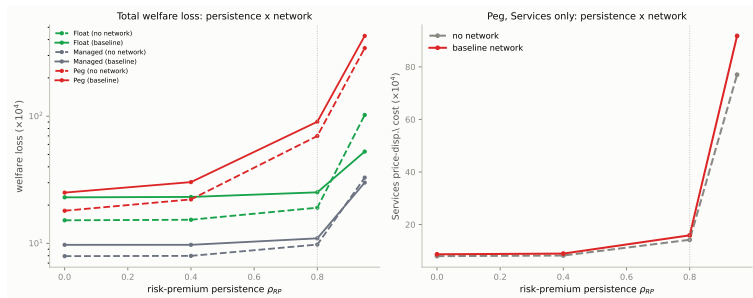
Does the network's effect scale *continuously* with how interconnected the economy is?



Denser network amplifies shock propagation for *everyone*, but **Peg** has no policy lever ($\phi_\pi, \phi_y = 0$) to lean against it, so the amplification passes straight into welfare loss; **Float/Managed** actively damp part of it with the interest rate — **gap widens** as $\rho \uparrow$

Persistence, Not Just Size: Why UIP Losses Are So Large

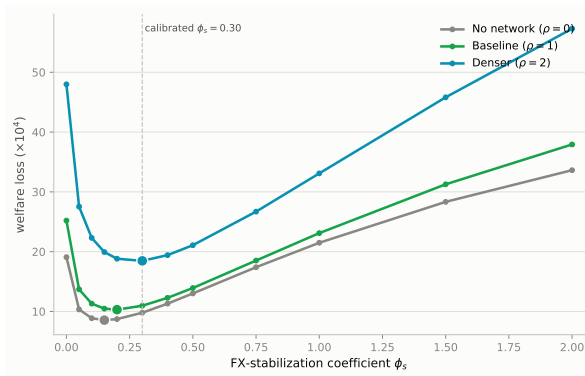
Christian: "*persistence of inflation leading to higher cost.*" The risk-premium/UIP shock's persistence ρ_{RP} (baseline 0.80) — crossed with network density — turns out to matter **far more convexly** than its size (previous risk-premium slide) or the network alone.



AR(1) variance scales as $\sigma^2/(1 - \rho_{RP}^2)$ — **mechanically convex**, not linear, in persistence. Moving ρ_{RP} from 0.40 $\rightarrow$ 0.80 raises Peg's loss only modestly ($18 \rightarrow 22 \rightarrow 70 \times 10^{-4}$ at $\rho=0$), but 0.80 $\rightarrow$ 0.95 **explodes it** to 343×10^{-4} — a 5x jump from a 0.15 change in persistence, not size

Does the Optimal Managed-Float Response Depend on the Network?

Christian's question: welfare loss changes a lot with network density — does the **welfare-minimizing** FX-stabilization coefficient ϕ_s change too, or just its magnitude?



Re-solves the managed-float rule $\hat{t}_t = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \log S_t$ over the ϕ_s grid at three network densities: **no network** ($\rho=0$), **baseline** ($\rho=1$), **denser** ($\rho=2$)

ϕ_s^* **rises with density**: 0.15 $\rightarrow$ 0.20 $\rightarrow$ 0.30 — denser network, more indirect cost-push (previous slide), so the welfare-minimizing rule leans harder against the exchange rate

How the Network Amplifies Shocks

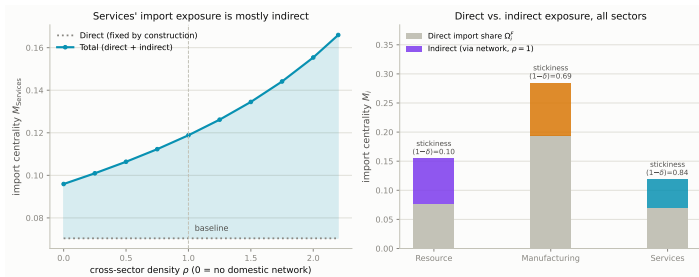
The network **dampens** the inflation peak but **amplifies** the output-gap peak — in every regime, for *both* shocks.

Shock	Regime	Inflation peak (Network ON ÷ Network OFF)	Output-gap peak
Import-price	Float	0.40×	2.77×
	Managed	0.61×	2.28×
	Peg	0.86×	3.15×
Risk-premium	Float	0.41×	2.57×
	Managed	0.57×	0.98×
	Peg	0.67×	2.16×

Note: $< 1\times$ = network shrinks the peak; $> 1\times$ = network amplifies it. (Peak IRF response over 20 quarters; exchange-rate peak itself barely moves with the network)

Why Services? Direct vs. Indirect Import Exposure

Services is the **stickiest** sector (Calvo reset prob. $\delta_3=0.16$) and has the **lowest direct** import share ($\Omega_3^F=0.070$) — yet it buys 5.8% of its cost from Manufacturing, the most import-intensive sector ($\Omega_2^F=0.195$).



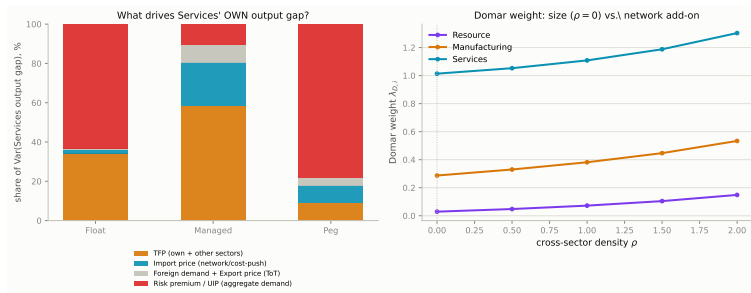
Total import centrality $M_i = [(I - \Omega^H)^{-1} \Omega^F \mathbf{1}]_i$ splits into a **direct** part (own Ω_i^F) and an **indirect** part inherited from suppliers — for Services, **41% of total exposure is indirect** (Resource 50%, Manuf. 32%)

Indirect exposure is **mechanically zero** at $\rho=0$ and grows with network density ρ — density doesn't change any sector's *direct* import share, only how much of *other* sectors' import content gets passed to it

Combined with stickiness: Services can't reprice quickly, so whatever cost-push it inherits shows up as persistent inflation, not a one-quarter blip — $\text{Corr}(\pi_3, \pi_3^{t-1}) \approx 0.82$, the most persistent inflation series in the model

Which Channel Actually Drives Services' Exposure?

Is it the import network, monetary mismatch, or just size? Dynare's own shock-by-shock variance decomposition of Services' **own** output gap says: mostly **neither** of the first two alone.



Left: the import-price/network channel (previous slide) is real but **secondary** for Services' output gap — 2% (Float) / 22% (Managed) / 9% (Peg) of its variance. **Risk-premium/UIP** — a pure aggregate, non-import shock — dominates instead: 63% (Float), 78% (Peg)

Right: Services' Domar weight $\lambda_{D,3}$ (its welfare-relevant "size") is 92% already present at $\rho=0$ — from its own consumption share $\beta_3^H=0.74$ plus heavy *within*-sector input reuse ($\Omega_{33}^H=0.27$) — the cross-sector network adds only $\approx 9\%$ more. Contrast Resource: network more than doubles its weight (60% of its total is network-added)

Synthesis — all three of Christian's hypotheses compound, none is "the" answer: size

Network Exposure & Sector Welfare Preferences

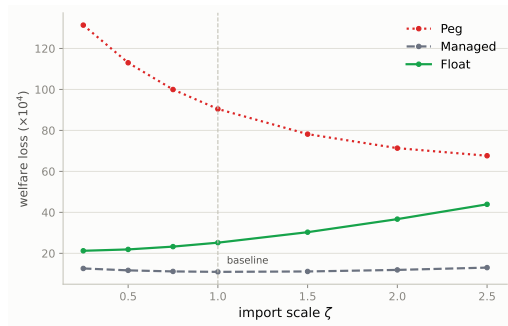
Each cell: Network ON / Network OFF.

Price-dispersion cost ($\times 10^4$)			
Sector	Float	Managed	Peg
Resource	0.05 / 0.03	0.04 / 0.02	0.27 / 0.27
Manuf.	3.87 / 2.23	1.92 / 0.94	4.28 / 5.32
Services	20.25 / 6.22	6.71 / 3.74	15.88 / 14.59

Var(own output gap) ($\times 10^4$)			
Sector	Float	Managed	Peg
Resource	0.13 / 0.02	0.14 / 0.04	0.20 / 0.12
Manuf.	0.58 / 0.16	0.38 / 0.08	1.98 / 0.23
Services	0.38 / 0.15	2.31 / 1.33	35.16 / 8.97

Every sector agrees: **Managed** < **Float** < **Peg** — Services' huge stake in avoiding Peg is mostly network-driven (output-gap cost 8.97 $\rightarrow$ 35.16 with the network on).

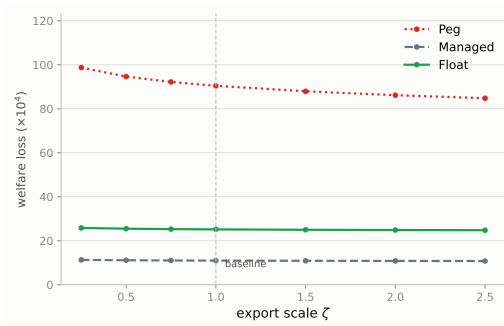
Import Openness



Managed < Float < Peg robust throughout

More import exposure means more $\Gamma \Delta e_t$ cost-push for **Float** (loss rises), but that channel is structurally zero under **Peg** ($\hat{s}_t \equiv 0$), so its loss instead falls via the labor-share renormalization — **gap converges**

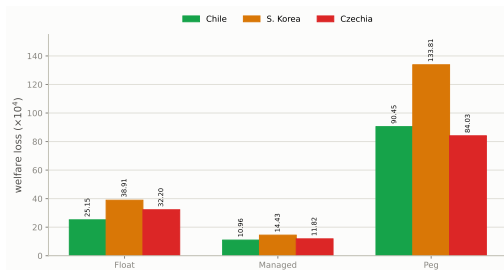
Export Openness



Managed < Float < Peg robust throughout

Exports work only through **demand** ($EX_{it} \rightarrow \tilde{y}_t$), never marginal cost — a pure buffer, so **all regimes improve**, and **Peg gains most** since it has no FX buffer of its own

Robustness: South Korea & Czechia

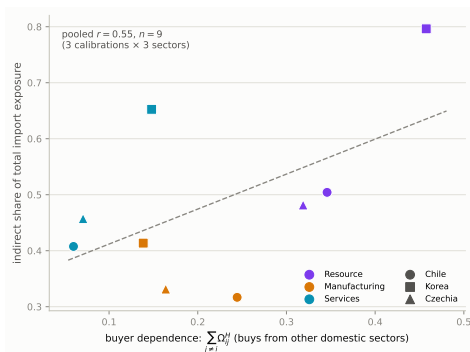


Managed < Float $\ll$ Peg in every calibration

Korea higher throughout: denser network & higher import exposure

Robustness: Is Indirect Exposure a General Pattern?

Adam's question: is "low direct import, high indirect import" (Services, two slides back) a general downstream-sector pattern, or a one-off? Pools **Chile, Korea, Czechia** × 3 sectors — 3 independent national IO tables, not the same data reused.



Honest answer: only a moderate, noisy relationship ($r=0.55$, $n=9$) between how much a sector sources domestically and how much of its import exposure is indirect — **with just 3 sectors this is a weak test**, not a proven general law

What IS robust across all three calibrations: Services sources the *least* domestically of any sector in every single calibration (lowest buyer-dependence, Chile/Korea/Czechia), yet still derives 41–65% of its import exposure indirectly — because what little it buys is

Conclusion

The network matters for severity and incidence, not the ranking: switching it on raises welfare losses in every regime and reallocates who bears them, but **Managed < Float < Peg either way**

Managed float wins because it dampens the FX pass-through channel without fully reimposing peg's loss of autonomy

Robust across three real-data calibrations (Chile, Korea, Czechia) and a genuine second-order solution